

Project: Introduction to Applying Optical Flow to Determine the Direction of Traffic

In this module, you have seen multiple techniques for detecting motion. Although the module project asks you to use optical flow, depending on your images, there is usually more than one correct approach.

In this module's project, you will apply what you have learned to detect moving cars from several frames of a video portraying a busy highway. This footage was taken using a camera on a tripod, so there is little need for camera stabilization.



In the first external MATLAB Grader tool, you will use optical flow to create a mask isolating the fastest-moving objects (cars) in each frame.



In the second external MATLAB Grader tool, you will apply this mask to calculate the velocity in the x-direction of each car and determine how many cars are moving in each direction.

Proceed to the next external MATLAB Grader tool to get started. You are encouraged to develop your code in MATLAB and copy it over to the Grader tool when you are ready. The images you are asked to perform optical flow on are included in the course files download.

If you get stuck, refer to the `applyingOpticalFlow.mlx` live script reading, which performs similar operations on moving cars, bicycles, and pedestrians.

Good luck!

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